



Walchand Institute of Technology, Solapur
(An Autonomous Institute)

Affiliated to
Punyashlok Ahilyadevi Holkar Solapur University,
Solapur

Choice Based Credit System (CBCS)

Structure and Syllabus
for
Final. Year. B.Tech. Information Technology
W.E.F. 2026-27

Mamale
HEAD
Information Technology Deptt.
W.I.T., Solapur.



WIT
Dean (Academics)

Department of Information Technology

Vision

- To be a frontier in Information Technology, to produce globally competent engineers with an aptitude for leadership and research, who will be instrumental in continuous socio-economic development.

Mission

- **M1:** To impart quality education in Information Technology in accordance with the needs of the society through blended mode.
- **M2:** To inculcate critical thinking and creativity for identifying various issues and to provide sustainable solutions by becoming a lifelong learner.
- **M3:** To enhance career opportunities through academia-industry interaction and research, while embodying professional ethics.



Department of Information Technology

Program Educational Objectives (PEOs)

- Graduates will exhibit strong fundamental knowledge and skills in the field of Information Technology to pursue successful professional careers, higher studies and research.
- Graduates will exhibit capabilities to understand and resolve the various issues through their problem-solving skills.
- Graduates will be sensitive to ethical, societal and environmental issues while serving at their professional work and society.

Knowledge and Attitude Profile (WK)

WK1	A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.
WK2	Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.
WK3	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
WK4	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.
WK5	Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.
WK6	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.
WK7	Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.
WK8	Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.
WK9	ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.



Program Outcomes (POs)	
PO 1	Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
PO 2	Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
PO 3	Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
PO 4	Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
PO 5	Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
PO 6	The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
PO 7	Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
PO 8	Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
PO 9	Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning difference
PO 10	Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
PO 11	Life-long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and



	emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)
Program Specific Outcomes (PSOs)	
● PSO 1: Graduates will be able to apply fundamentals of mathematics, algorithms and computational systems to Information technology. ● PSO 2: Students will be able to provide a solution to the problem in the areas of Networking, Database management, System Software, Web Technology, Information Security and Thrust areas. ● PSO 3: Students will be able to design and develop IT solution for societal problem/s while encouraging usage of Free and Open-Source Software (FOSS).	



Department of Information Technology

Legends Used

L	Lecture Hours / week
T	Tutorial Hours / week
P	Practical Hours / week
FA	Formative Assessment
SA	Summative Assessment
ESE	End Semester Examination
ISE	In Semester Evaluation
ICA	Internal Continuous Assessment
POE	Practical and Oral Exam
OE	Oral Exam
MOOC	Massive Open Online Course
HSS	Humanity and Social Science
NPTEL	National Program on Technology Enhanced Learning
F.Y.	First Year
S.Y.	Second Year
T.Y.	Third Year
B. Tech.	Bachelor of Technology



Department of Information Technology

Course Code Format

2	2	I	T	U/P	2	C	C	1	T/L
Year of Syllabus revision		Program Code		U-Under Graduate P-Post Graduate	Semester No./ Year 1/2/3/...8	Course Type		Course Serial No 1-9	T-Theory, L-Lab session A-Tutorial P-Programmin g

Program Code

IT	Information Technology
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Course Type

BS	Basic Science
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ES	Engineering Science
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HU	Humanities & Social Science
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MC	Mandatory Course
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CC	Core Compulsory Course
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SN*	Self-Learning <i>N* indicates the serial number of electives offered in the respective category</i>
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EN*	Core Elective <i>N* indicates the serial number of electives offered in the respective category</i>
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ON*	Open Elective <i>N* indicates the serial number of electives offered in the respective category</i>
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SK	Skill Based Course
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SM	Seminar
MP	Mini project
PR	Project
IN	Internship

Sample Course Code	
22ITU7CC1T	Cyber Security



Department of Information Technology

B. Tech. Semester –VII

Course Code	Theory Course Name	Engagement Hours			Credits	FA		SA		Total
		L	T	P		ESE	ISE	ICA		
23ITU7CC1T	Cyber Security	2	-	-	2	60	40			100
23ITU7CC2T	Agentic AI	3	-	-	3	60	40			100
23ITU7EN*3T	Program Elective IV	3	-	-	3	60	40			100
23ITU7EN*4T	Program Elective V	2	-	-	2	60	40			100
23##U7MD5T	Multidisciplinary Minor - V	3	-	-	3	60	40			100
23##U7MD5A	Multidisciplinary Minor - V Tutorial		1		1			25		25
Sub Total		13	1		14	300	200	25		525
Course Code	Laboratory Course Name									
						ESE				
						POE	OE			
23ITU7CC1L	Cyber Security	-	-	2	1	-	-	-	25	25
23ITU7CC2L	Agentic AI	-	-	2	1	25	-	-	25	50
23ITU7EN*3L	Program Elective IV	-	-	2	1	-	-	-	25	25
23ITU7EN*4L	Program Elective V	-	-	2	1	-	-	-	25	25
23ITU7PR6L	Project Phase I	-	-	8	4	50	-	-	100	150
Sub Total				16	8	75			200	275
Grand Total		13	1	16	22	375		200	225	800



Department of Information Technology

B. Tech. Semester –VIII

Course Code	Theory Course Name	Engagement Hours			Credits	FA	SA		Total
		L	T	P		ESE	ISE	ICA	
23CMU8RN*1T	Research Methodology & IPR / MOOC#	-	-	-	2	50			50
23ITU8EN*2T	Program Elective VI / MOOC#	-	-	-	2	50			50
Sub Total		-	-	-	4	100			100
Course Code	Laboratory Course Name								
						ESE			
						POE	OE		
23ITU8IN3L	Internship	-	-	20	10	50		100	150
23ITU8PR4L	Project Phase II	-	-	4	2	50		100	150
Sub Total		-	-	24	12	100		200	300
Grand Total		-	-	24	16	200		200	400

Note:

- N* indicates the serial number of electives offered in the respective category
- # List of Courses will be provided by the BoS.

List of Program Elective Course -IV (Semester –VII)

Course Code	Course title
23ITU7E13T	Deep Learning
23ITU7E23T	Quantum Computing
23ITU7E33T	Business Intelligence

List of Program Elective Course – V (Semester –VII)

Course Code	Course title
23ITU7E14T	Generative AI
23ITU7E24T	Edge Computing
23ITU7E34T	Prompt Engineering



List of Program Elective Course -VI (Semester –VIII)	
Course Code	Course title
23ITU8E12T	Software Testing and Quality Assurance
23ITU8E22T	Augmented Reality & Virtual Reality

For Courses offered during Semester VIII for A.Y. 2026-27

Sr. No.	Course Code	Option -I	Option II
1	23CMU8RN*1T	‘Research Methodology & IPR’ course offered by the department during Semester VIII in Self Learning / Online mode and successfully passing ESE for the same. 23CMU8R11T – Self Learning Mode	MOOC Certification* 23CMU8R21T – For MOOC (Option I) 23CMU8R31T – For MOOC (Option II)
2	23ITU8EN*2T	Program Elective VI’ course offered by the department during Semester VIII in Self Learning / Online mode and successfully passing ESE for the same. Program Elective Course VI options: • 23ITU8E12T - Software Testing and Quality Assurance • 23ITU8E22T- Augmented Reality & Virtual Reality	MOOC Certification* 23ITU8E32T – MOOC (Option I) 23ITU8E42T – MOOC (Option II) 23ITU8E52T – MOOC (Option III)

- If students have not completed the MOOC in Semester V or VI, then students have to complete this MOOC in the Semester VII or VIII. In Semester VIII, the marks earned by the students in final assessment of this MOOC will be appropriately scaled and transferred for 23CMU8RN*1T and / or 23ITU8EN*2T along with the required credits. The list of approved MOOCs will be finalized and released by Board of Studies before the commencement of the semester as per the available courses on MOOC platform.
- If students do not pass the MOOC examination of 23CMU8RN*1T or 23ITU8EN*2T before commencement of End Semester Examination (ESE) of Semester VIII, then they must appear for the institute-conducted ESE for courses (mentioned in option I) by department during Semester VIII.



**WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR****(An Autonomous Institute)****Final Year B.Tech (Information Technology), Semester-VII****23ITU7CC1T: Cyber Security**

Teaching Scheme		Examination Scheme	
Lectures	2 Hours/week	ESE	60 Marks
Practical	2 Hours/week	ISE	40 Marks
Credits	3	ICA	25 Marks
Introduction:			
This course introduces students to the fundamental concepts of cyber security, cybercrime, cyber offenses, cyber-attack methods, phishing, identity theft, and computer forensics. It develops an understanding of security principles, common cyber threats, legal and social aspects of cybercrime, and basic forensic practices required to protect digital systems and investigate cyber incidents. The course also encourages students to explore practical problem-solving through assignment-based learning.			
Course Prerequisite: A basic understanding of computers, networking, and Internet usage is recommended to effectively learn cyber security concepts.			
Course Objectives:			
1. To introduce the basic principles of cyber security and cryptographic techniques. 2. To provide knowledge of cybercrime, cyber offenses, and cyber attack strategies. 3. To develop awareness about phishing, identity theft, and cybercrime tools and methods. 4. To familiarize students with computer forensics and digital investigation procedures.			
Course Outcomes:			
After completing the course, students will be able to 1. Demonstrate cyber security principles, security mechanisms, and classical encryption techniques. 2. Describe types of cybercrime, cyber offenses, and their legal perspectives. 3. Illustrate phishing, identity theft, and methods used in cyber-attacks. 4. Use computer forensics to handle digital evidence and cyber investigations.			
Unit – I	Security Fundamentals		5 Hours



Computer Security Concepts, OSI Security Architecture, Security Attacks, Security Services, Security Mechanisms, A Model for Network Security, Classical Encryption Techniques including Symmetric Cipher Model and Cryptography, Substitution Techniques such as Caesar Cipher, Monoalphabetic Cipher, Playfair Cipher, Hill Cipher, Polyalphabetic Cipher, Transposition Techniques, Rotor Machines, and Steganography.		
Unit – II	Introduction to Cybercrime	5 Hours
Introduction, Cybercrime: Definition and Origins of the Word, Cybercrime and Information Security, Who are Cybercriminals? Classifications of Cybercrimes, Cybercrimes: An Indian Perspective, Hacking and the Indian Laws, A Global Perspective on Cybercrimes.		
Unit – III	Cyber offenses	5 Hours
Cyber offenses: How Criminals Plan Them: Introduction, How Criminals Plan the Attacks, Social Engineering, Cyberstalking, Cybercafe and Cybercrimes. Botnets: The Fuel for Cybercrime, Attack Vector.		
Unit – IV	Tools and Methods Used in Cybercrime	5 Hours
Introduction, Proxy Servers and Anonymizers, Phishing, Password Cracking, Keyloggers and Spywares, Virus and Worms, Trojan Horses and Backdoors, Steganography, DoS and DDOS Attacks, Attacks on Wireless networks		
Unit – V	Phishing and Identity Theft	5 Hours
Phishing and Identity Theft: Introduction, Methods of Phishing, Phishing Techniques, Spear Phishing, Types of Phishing Scams, Phishing Toolkits and Spy Phishing, Phishing Countermeasures, Identity Theft (ID Theft)		
Unit – VI	Understanding Computer Forensics	5 Hours
Introduction, Historical Background of Cyberforensics, Digital Forensics Science, The Need for Computer Forensics, Cyberforensics and Digital Evidence, Digital Forensics Life cycle, Chain of Custody Concept, Network Forensics.		
ICA consists of minimum 10 assignments requiring students to analyze cyber security concepts and real-world cybercrime scenarios based on the following topics: 1. Security Fundamentals 2. Classical Cryptography Techniques 3. Cybercrime Terminologies and Classification 4. Cybercrime: Indian and Global Perspective 5. Cyber Offenses and Attack Planning		



6. Botnets and Attack Vectors 7. Tools and Methods Used in Cybercrime 8. Phishing and Identity Theft 9. Cyber Attack Countermeasures 10. Computer Forensics and Digital Evidence
Text Books
1. Sunit Belapure, Nina Godbole, “Cyber Security: Understanding Cyber Crimes, Computer Forensics and Legal Perspectives”, Wiley India Pvt Ltd, ISBN: 978-81- 265-21791, 2011, First Edition (Reprinted 2018).
Reference Books
1. Thomas J. Mowbray, “Cybersecurity: Managing Systems, Conducting Testing, and Investigations”, John Wiley & Sons, ISBN: 978-1-118-69711-5, 2014.
e-Resources
1. https://onlinecourses.swayam2.ac.in/e-learning/preview/nou19_cs08



**WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR****(An Autonomous Institute)****Final Year B.Tech. (Information Technology), Semester-VII****23ITU7CC2T: Agentic AI****Teaching Scheme****Examination Scheme**

Lectures	3 Hours/week	ESE	60 Marks
Practical	2 Hours/week	ISE	40 Marks
Credits	4	ICA	25 Marks
		POE	25 Marks

Introduction:

This course provides a foundational understanding of Agentic AI systems and demonstrates how intelligent agents can autonomously plan, reason, and perform tasks to solve real world problems.

Course Prerequisite:

Programming, probability, statistics, GenAI.

Course Objectives:

1. To explain the fundamentals of AI agents and LLM-based systems.
2. To analyze the design and architecture of agentic and multi-agent systems.
3. To develop skills for building and deploying Agentic AI applications.
4. To apply Agentic AI concepts to solve real-world problems.

Course Outcomes:

After completing the course, students will be able to

1. Demonstrate the foundational concepts of AI agents and LLM-based systems.
2. Analyse the design of AI agents and multi agent systems.
3. Build scalable and efficient agentic AI applications.
4. Develop problem-solving skills by designing agentic solutions for various domains.

Unit – I**Introduction to Agentic AI****8 Hours**

What is Artificial Intelligence, Evolution of AI Systems, Traditional AI vs. Agent-Based, Components of an intelligent agent (environment, perception, action), Types of agents (reactive,



deliberative, hybrid), Real-world applications of AI agents.		
Unit – II	Foundations for Building AI Agents	9 Hours
Basics of programming for AI (Python overview), Role of probability and statistics in decision-making, Introduction to machine learning concepts APIs, tools, and data handling for agents, Large Language Models like RAG and Hugging Face etc.		
Unit – III	Agent Design and Multi-Agent Systems	8 Hours
Transition from workflows to AI agents, Agent architectures (ReAct, tool-using agents), Multi-agent collaboration and orchestration, Prompting strategies for agents, Task decomposition and supervision.		
Unit – IV	Advanced Agent Systems and Deployment	12 Hours
Multimodal AI agents (text, image, audio, code), Reasoning LLMs and computer-use agents, Fine-tuning and optimization techniques, Production deployment and scaling, Cost, privacy, AI Ethics, AI governing laws and real-world constraints.		
Unit – V	Case Studies	8 Hours
Customer support agent, personal assistant agent, legal agent, advertising agent, finance agent, education agent		
The ICA consists of a minimum of 8 assignments, requiring students to study, analyse, and implement Agentic AI case studies for real-world problems, use cases, or scenarios, to design and build applications where AI acts as an enhancement layer, improving usability, automation, and decision-making. 1. Identify agentic AI Case Study – Understanding real-world applications and identifying problem statements 2. Analysis of LLM-Based Solutions – Studying how LLMs are used in existing case studies 3. Data Handling and Knowledge Integration – Exploring datasets and retrieval mechanisms used in case studies 4. Single-Agent Case Study Implementation – Implementing a basic AI agent for a selected use case (e.g., chatbot or assistant) 5. Tool-Integrated Case Study – Developing agents that interact with external tools/APIs in real scenarios 6. Agent Memory and Workflow Analysis – Implementing context handling and multi-step workflows in case studies 7. Multi-Agent Case Study Implementation – Designing and implementing collaborative agent systems for complex problems 8. Final Case Study Project – End-to-end implementation of a domain-based Agentic AI solution including analysis, design, implementation, and evaluation		
Text Books		



1. *Building Applications with AI Agents*, Michael Albada, O'Reilly, 2026
2. *Building Agentic AI: Workflows, Fine-Tuning, Optimization, and Deployment*, Sinan Ozdemir, Pearson AI, 2026.

Reference Books

1. *AI Agents*, Avi Chawla and Akshay Pachaar, Daily Dose of Data Science, 2025.

E-Resources

1. LangChain Documentation
2. OpenAI API Documentation



**WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR****(An Autonomous Institute)****Final Year B.Tech. (Information Technology), Semester-VII****23ITU7E13T– Program Elective IV: Deep Learning****Teaching Scheme****Examination Scheme**

Lectures	3 Hours/week	ESE	60 Marks
Practical	2 Hours/week	ISE	40 Marks
Credits	4	ICA	25 Marks

Introduction:

This course provides a fundamental understanding of Deep Learning techniques and neural network architectures used to solve complex problems in areas such as image processing, speech recognition, and computer vision. It also introduces modern tools like TensorFlow, Keras, and PyTorch for implementing deep learning models.

Course Prerequisite:

Basic knowledge of Machine Learning concepts, programming (preferably Python), and fundamentals of mathematics such as linear algebra, calculus, probability, and statistics.

Course Objectives:

1. To compare between Machine Learning and Deep Learning
2. To learn the basics of neural networks and their needs.
3. To learn and differentiate among various deep learning models.
4. To apply Deep learning techniques to solve various real time problems.

Course Outcomes:

Student will be able to-

1. Differentiate between Machine Learning and Deep Learning by comparing their concepts, architectures, and application areas.
2. Demonstrate the structure and working of artificial neural networks, including neurons, activation functions, and learning mechanisms.
3. Analyze and compare different deep learning models such as Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN) based on their architecture and applications.
4. Implement and evaluate deep learning techniques to solve real-world problems using appropriate tools and performance metrics.



Unit – I	Foundations of Deep learning	7 Hours
What is machine learning and deep learning? Supervised and Unsupervised Learning, Limitations of machine learning, History of deep learning, Advantages and challenges of deep learning. Understanding how deep learning works, Common Architectural Principles of Deep Network, Architecture Design, Applications of Deep learning, Introduction, and use of popular industry tools such as Tensorflow, Keras, PyTorch.		
Unit – II	Deep Neural Networks (DNNs)	8 Hours
Introduction to Neural Networks: The Biological Neuron, The Perceptron, Multilayer Feed-Forward Networks, Training Neural Networks: Backpropagation and Forward propagation Activation Functions: Linear, Sigmoid, Tanh, Softmax, Rectified Linear, Loss Functions: Loss Function Notation, Loss Functions for Regression, Loss Functions for Classification, Hyperparameters: Learning Rate, Regularization, Momentum, Sparsity, Deep Feedforward Networks – Example of Ex OR, Hidden Units, cost functions, error backpropagation, Gradient-Based Learning, Implementing Gradient Descent, vanishing and exploding gradient descent, Deep Learning with Pytorch, Jupyter, colab.		
Unit – III	Convolution Neural Network (CNN)	8 Hours
Introduction, CNN architecture overview, The Basic Structure of a Convolution Network-Padding, Strides, Typical Settings, the ReLU layer, Pooling, Fully Connected Layers, The Interleaving between Layers, Local Response Normalization, Training a Convolution Network		
Unit – IV	Recurrent Neural Networks	8 Hours
Recurrent and Recursive Nets: Recurrent Neural Networks, Bidirectional RNN, Encoder Decoder Architectures, Introduction to Long Short-Term Memory (LSTM) and Temporal Dependencies, Gated Recurrent Units (GRUs), Applications of RNN in Real World- Image Captioning and Time Series Forecasting and Prediction		
Unit – V	Autoencoders	6 Hours
Under complete Auto encoders, Regularized Autoencoders-Sparse Autoencoders, Stochastic Encoders and Decoders, Applications of Autoencoders, Introduction to transfer learning		
Unit – VI	Applications of Deep Learning	8 Hours
Computer vision, Object Detection using YOLO model, Introduction to deep generative model, Boltzmann Machine, Deep Belief Networks, Generative adversarial network (GAN), Discriminator network, Generator network, types of GAN, Applications of GAN networks.		
Internal Continuous Assessment (ICA): ICA shall include at least eight of the following:		



1. Implementation of a deep learning model with MNIST data using Caffe2
2. Implementation of sentiment analysis using LSTM using Amazon Alexa reviews Dataset.
3. Implementation of Airline sentiment analysis using LSTM using Tweeter reviews Dataset.
4. Implementation of CNN for flower Image classification
5. Build a MLP model to demonstrate the effect of Batch Normalization using Keras.
6. Implementation of CNN for Image classification using MNIST dataset
7. Implementation of CNN for Hand written digit recognition using MNIST dataset
8. Transfer learning using ResNet50 for image recognition
9. Build a Keras classification Model on the Diabetes Dataset
10. Implementation of CNN model for animal image classification using Fashion MNIST
11. Implementation of Autoencoder techniques
12. Application of Generative models

Text Books

1. Goodfellow, I., Bengio, Y. Courville, A, "Deep Learning", MIT Press, 2016.
2. Charu Agarwal, "Neural Networks and deep learning", A Textbook, Springer International Publishing AG, part of Springer Nature 2018 , ISBN 978-3-319-94462-3
3. Deep Learning - A Practical Approach by Rajiv Chopra, Khana Publications, 2018, ISBN: 9789386173-41-6

Reference Books

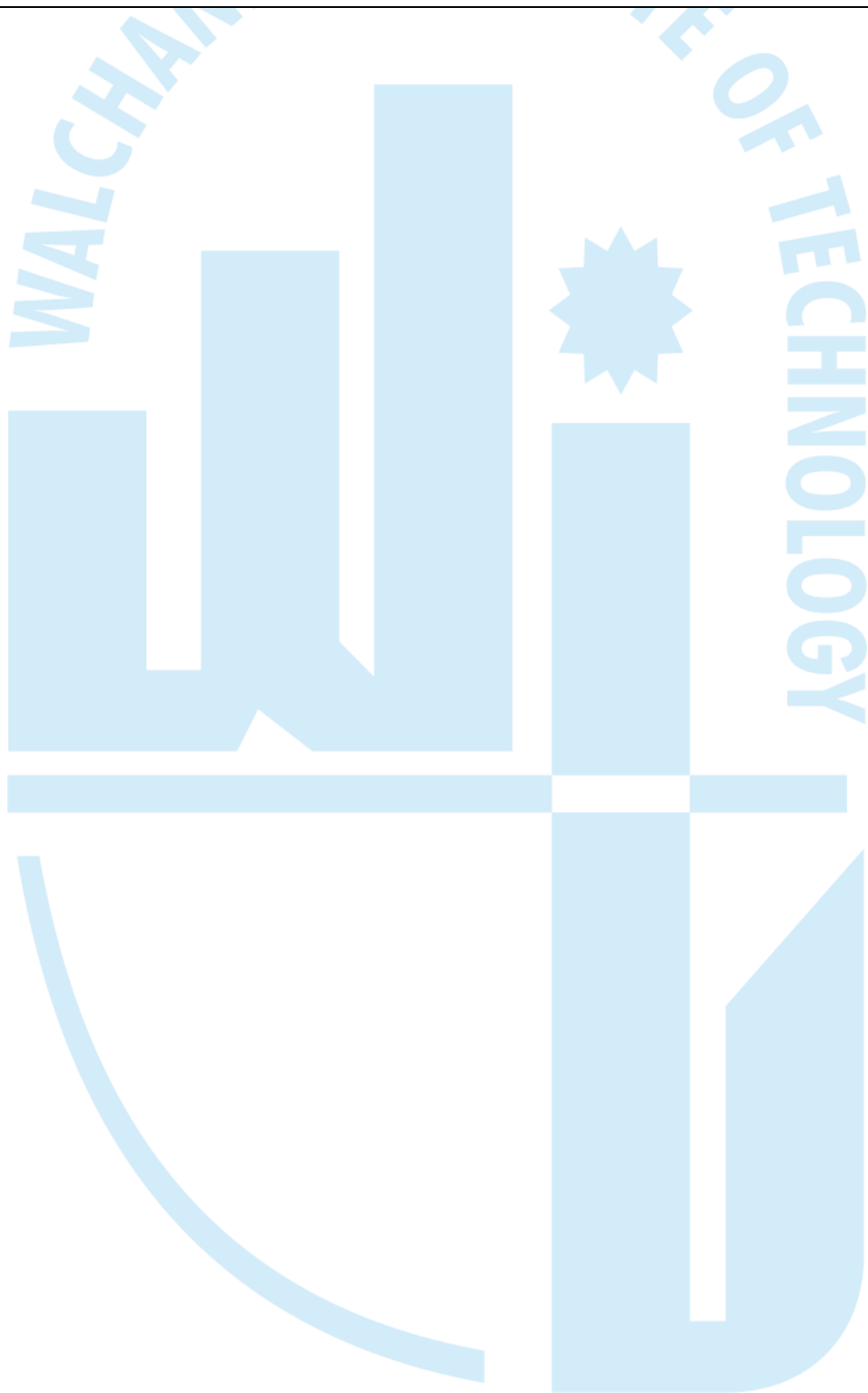
1. Josh Patterson, Adam Gibson "Deep Learning: A Practitioner's Approach", O'Reilly Media, 2017

E-Resources

1. <http://csis.pace.edu/ctappert/cs855-18fall/DeepLearningPractitionersApproach.pdf>
2. https://www.dkriesel.com/_media/science/neuronale-netze-en-zeta2-1col-dkrieselcom.pdf
3. MOOC Courses Links: <https://www.my-mooc.com/en/categorie/deep-learning>
4. Michael Nielsen, "Neural Networks and Deep Learning", Online book, 2016 (<http://neuralnetworksanddeeplearning.com/>)
5. Deep Learning for Visual Computing https://onlinecourses.nptel.ac.in/noc22_ee54
6. Deep Learning - IIT Kharagpur https://onlinecourses.nptel.ac.in/noc22_cs22



7. Deep Learning - IIT Ropar https://onlinecourses.nptel.ac.in/noc22_cs35/
8. Introduction to Deep Learning: <https://www.coursera.org/learn/introduction-to-deeplearning-boulder>
9. Deep Learning Specialization : <https://www.coursera.org/specializations/deep-learning>



**WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR****(An Autonomous Institute)****Final Year B.Tech. (Information Technology), Semester-VII****23ITU7E23T - Program Elective IV: Quantum Computing**

Teaching Scheme		Examination Scheme	
Lectures	3 Hours/week	ESE	60 Marks
Practical	2 Hours/week	ISE	40 Marks
Credits	4	ICA	25 Marks
Introduction:			
This course provides a foundational understanding of quantum computing concepts, architectures, and algorithms. It equips engineering students with the knowledge required to analyse and develop quantum solutions for complex computational problems.			
Course Prerequisite:			
Students should have basic knowledge of linear algebra, probability theory, and classical algorithms. Familiarity with Python programming and fundamentals of computer architecture is recommended.			
Course Objectives:			
1. To introduce fundamental principles of quantum mechanics relevant to quantum computation. 2. To develop the ability to Analyze and design quantum circuits using quantum gates and registers. 3. To study quantum algorithms and understand their computational advantages over classical algorithms. 4. To provide hands-on exposure to quantum programming platforms and emerging real-world applications.			
Course Outcomes:			
After completing the course, students will be able to 1. Demonstrate the basic principles of quantum computing and quantum mechanics. 2. Design and analyze simple quantum circuits and gates. 3. Implement basic quantum algorithms using quantum programming tools. 4. Identify applications and future challenges of quantum computing.			
Unit – I	Foundations of Quantum Mechanics		8 Hours



Limitations of classical computation and motivation for quantum computing, Basic postulates of quantum mechanics for computation, Qubits and Quantum States, Dirac Notation , Superposition and Normalization, Measurement and Probability, Bloch Sphere Representation		
Unit – II	Quantum Gates and Circuit Model	8 Hours
Quantum Registers, Single Qubit Gates: X, Y, Z, H, Phase, Multi-Qubit Gates: CNOT, Toffoli, SWAP Quantum Circuit Model, Reversible Computing, Quantum Entanglement, No-Cloning Theorem		
Unit – III	Quantum Algorithms	8 Hours
Classical vs Quantum Algorithmic Speedup, Deutsch–Jozsa Algorithm, Grover’s Search Algorithm, Shor’s Factoring Algorithm, Quantum Fourier Transform (QFT), Applications in Cryptography		
Unit – IV	Quantum Error Correction & Noise	8 Hours
Decoherence and Quantum Noise, Quantum Errors: Bit-Flip, Phase-Flip, Quantum Error Correction Codes, Shor Code and Steane Code, Fault-Tolerant Quantum Computing, Physical limitations of Quantum Systems.		
Unit – V	Quantum Programming & Platforms	7 Hours
Quantum Programming Concepts, Introduction to Qiskit / Cirq, Writing Quantum Programs, Quantum Simulators vs Real Quantum Hardware, Cloud-based Quantum Platforms, Case Studies: Optimization, AI, Chemistry		
Unit – VI	Applications & Future Directions	6 Hours
Quantum Cryptography (BB84 Protocol), Quantum Key Distribution (QKD), Quantum Machine Learning, Quantum Computing in Drug Discovery, Current Challenges and Research Trends, Future Scope and Industrial Applications		
ICA consists assignments requiring students to: 1. Explain fundamental concepts of quantum computing with suitable examples. 2. Compare classical bits and qubits using diagrams and mathematical representation. 3. Represent single-qubit and multi-qubit states using Dirac notation. 4. Analyze quantum gates and construct simple quantum circuits. 5. Simulate basic quantum circuits using a quantum programming framework. 6. Implement and analyze Deutsch–Jozsa or Grover’s algorithm.		



7. Study quantum entanglement and demonstrate it using circuit simulation.
8. Investigate quantum noise and error correction techniques through case studies.
9. Prepare a technical report on applications of quantum computing in cryptography, AI, or optimization.
10. Develop a mini-project or presentation using a cloud-based quantum computing platform.

Text Books

1. Quantum Computation and Quantum Information – Michael A. Nielsen, Isaac L. Chuang, 10th Anniversary Edition, Cambridge University Press.
2. An Introduction to Quantum Computing – Phillip Kaye, Raymond Laflamme, Michele Mosca, First Edition, Oxford University Press.

Reference Books

1. Quantum Computing: A Gentle Introduction – Eleanor Rieffel, Wolfgang Polak, First Edition, MIT Press.
2. Qiskit Documentation – IBM Quantum Team, Qiskit Official Documentation, IBM.

e-Resources

UNIT I – Fundamentals of Quantum Mechanics

- <https://www.geeksforgeeks.org/blogs/introduction-quantum-computing/>
- <https://quantum.country/qcvc>

UNIT II – Quantum Gates and Circuits

- <https://monitsharma.github.io/Learn-Quantum-Computing-with-Qiskit/>
- <https://qiskit.org/textbook/ch-gates/what-is-a-quantum-gate.html>
- <https://quantumcomputinguk.org/gates-circuits-explained/>

UNIT III – Quantum Algorithms

- <https://quantum.cloud.ibm.com/learning/modules/computer-science/grovers?authuser=>
- <https://arxiv.org/abs/1903.04359>

UNIT IV – Quantum Error Correction & Noise

- https://qosf.org/learn_quantum
- <https://quantumcomputinguk.org/quantum-error-correction-explained>

Unit V – Quantum Programming & Platforms

- <https://quantumai.google/cirq/start/basics>
- https://learn.microsoft.com/en-us/azure/quantum/qsharp-overview?utm_source=



UNIT VI- Applications & Future Directions

- <https://quantumcountry.com/>
- <https://arxiv.org/abs/2111.15352>
- <https://qiskit.org/advocacy/applications>



**WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR****(An Autonomous Institute)****Final Year B.Tech. (Information Technology), Semester-VII****23ITU7E33T – Program Elective IV: Business Intelligence**

Teaching Scheme		Examination Scheme	
Lectures	3 Hours/week	ESE	60 Marks
Practical	2 Hours/week	ISE	40 Marks
Credits	4	ICA	25 Marks
Introduction:			
Business Intelligence turns raw data into useful information for better decision-making. This subject introduces BI systems, data warehouses, OLAP, and basic data mining techniques like classification and clustering. It focuses on analyzing large datasets and applying these concepts to real-world business problems.			
Course Prerequisite:			
Basics of Programming Python, SQL			
Course Objectives:			
1. To introduce fundamental concepts and organizational relevance of Business Intelligence and Analytics. 2. To develop the ability to analyze organizational problems using Decision Support Systems. 3. To provide a practical understanding of data warehousing, OLAP, and multidimensional data analysis. 4. To introduce data mining techniques for knowledge discovery. 5. To familiarize learners with BI and data mining applications across industries and emerging trends.			
Course Outcomes:			
After completing the course, students will be able to			
1. Describe the fundamental concepts and organizational role of Business Intelligence and Analytics. 2. Analyze decision-making problems and illustrate the use of Decision Support Systems in organizations. 3. Apply data warehousing and OLAP concepts for multidimensional data analysis. 4. Apply data mining techniques for classification and clustering tasks. 5. Demonstrate BI and data mining applications across different industries and emerging trends.			
Unit – I	Introduction to Business Intelligence & Analytics		4 Hours
Introduction to Business Intelligence (BI), Evolution of BI, Data, Information, Knowledge, and			



Wisdom (DIKW), BI architecture and components, Role of BI in organizations, BI vs traditional reporting systems, Benefits and challenges of BI, Overview of BI tools and platforms		
Unit – II	Decision Support Systems (DSS)	8 Hours
Introduction to Decision Support Systems, Types of decisions: Structured, Semi-structured, Unstructured, Components of DSS, DSS architecture, Model-driven, Data-driven, Knowledge-driven DSS, Role of BI in decision making, Executive Information Systems (EIS), Group Decision Support Systems (GDSS)		
Unit – III	Data Warehouse and OLAP	6 Hours
Introduction to Data Warehousing, Operational Database vs Data Warehouse, Characteristics of Data Warehouse, Data Warehouse architecture, ETL (Extract, Transform, Load) process, Metadata and data marts, Star schema and Snowflake schema, Introduction to OLAP, Multidimensional data models		
Unit – IV	Data Mining	7 Hours
Introduction to Data Mining, Data mining process (KDD), Data preprocessing techniques, Data mining tasks and functionalities, Classification: Decision Trees, Naïve Bayes, k-Nearest Neighbor (k-NN), Clustering: Partitioning methods (k-means), Hierarchical clustering, Comparison of classification and clustering, Applications of data mining		
Unit – V	BI and Data Mining Applications	5 Hours
BI and Data Mining in Banking and Finance, Retail and Marketing, Healthcare, Manufacturing, Telecommunications, E-commerce, Fraud detection, Customer relationship management (CRM), Recommendation systems, Web and social media analytics, Future trends in BI and Data Mining		
ICA consists of a minimum of 10 assignments requiring students to develop machine learning applications for real-world problems/use-cases/scenarios based on the following topics:		
1. Study of BI tools (Power BI / Tableau / Google Data Studio) 2. Creating simple dashboards using sample datasets 3. Identifying BI requirements for a given business case 4. Design a DSS model for a business scenario 5. Implement simple decision models using spreadsheets 6. Case study analysis of DSS in organizations 7. Design a star and snowflake schema for a given problem		



8. Implement ETL using tools or SQL
9. Perform OLAP operations using BI tools
10. Data preprocessing using Python / WEKA / R
11. Implement classification algorithms on datasets
12. Implement clustering algorithms and analyze results
13. Visualization of mined patterns
14. Mini project using BI or Data Mining

Tools: Power BI / Tableau, MySQL / PostgreSQL, Python (Pandas, Scikit-learn), WEKA / R

Text Books

1. Data mining and Business Intelligence – S. K. Shinde and Uddagiri Chandrasekhar, Dreamtech Publication
2. Business Intelligence: Data Mining and Optimization for Decision Making- Carlo Vercellis, Wiley India publication

Reference Book

1. Business Intelligence & Analytics - Efraim Turban, Ramesh Sharda, Dursun Delen





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR

(An Autonomous Institute)

Final Year B.Tech. (Information Technology), Semester-VII

23ITU7E14T- Program Elective V - Generative AI

Teaching Scheme		Examination Scheme	
Lectures	2 Hours/week	ESE	60 Marks
Practical	2 Hours/week	ISE	40 Marks
Credits	3	ICA	25 Marks
Introduction:			
This syllabus provides a comprehensive exploration of Generative AI, starting with an introduction to its history, evolution, and ethical considerations. It covers the mathematical foundations critical for understanding generative models, including probability, statistics, and information theory. The course delves into machine learning basics, discussing key frameworks like TensorFlow and PyTorch. Advanced units focus on various generative models such as GANs, VAEs, and large language models, and their applications in natural language processing, computer vision, and other domains, while also examining emerging trends and ethical implications			
Course Prerequisite:			
Basics of Programming, Probability, Statistics			
Course Objectives:			
1. To introduce the Fundamentals of Generative AI 2. To develop a solid grasp of the mathematical concepts essential for generative AI 3. To introduce Key Machine Learning Techniques . 4. To solve Real-World Problems using Generative AI.			
Course Outcomes:			
After completing the course, students will be able to 1. Demonstrate a thorough understanding of the fundamental concepts and ethical considerations of Gen AI. 2. Acquire proficiency in the mathematical foundations required for Gen AI. 3. Implement various Gen AI models. 4. Apply Gen AI techniques across various domains.			



Unit – I	Introduction to Generative AI	3 Hours
Overview of Generative AI, History and evolution of Generative AI, Applications and use cases, Ethical considerations and challenges of Generative AI		
Unit – II	Mathematical Foundations	5 Hours
Overview of Probability and statistics for AI, Role of Probability in Generative AI Statistics: Understanding Data Distributions, variables, Bayesian Inference in Generative Models, Estimation and Hypothesis Testing, Overview of Linear algebra and calculus, Information theory		
Unit – III	Machine Learning Basics	5 Hours
Introduction to Supervised and unsupervised learning algorithms, Different types of Supervised and unsupervised algorithms, Introduction to Neural networks and deep learning, Evaluation metrics and model validation, Introduction to key frameworks (TensorFlow, PyTorch)		
Unit – IV	Generative Models	5 Hours
Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), Generative vs. Discriminative Models, Flow-based models, Energy-based models, Large Language Models and its examples.		
Unit – V	Natural Language Processing (NLP) with Generative AI	5 Hours
Introduction to Text generation models, Transformer architecture (GPT-Generative Pre -trained transference, BERT- Bidirectional Encoder Representations from Transformers), Lang chain framework basics, Sequence-to-sequence models, Applications in language translation, summarization, and dialogue systems		
Unit – VI	Applications and Future Trends	7 Hours
Generative AI in computer vision (image generation, style transfer), Generative AI in healthcare, finance, and entertainment, Emerging trends and future research directions Ethical AI and responsible innovation.		
ICA consists of minimum 8-10 assignments requiring students to applications for real world problem/use- case/scenario based on the following topics:		
1. Exploring Generative AI Tools		



2. Statistical Distributions in AI
3. Implementation of Bayesian Inference
4. Supervised Learning Implementation
5. Unsupervised Learning - Applying Clustering Techniques
6. Neural Network Implementation
7. Text Generation using Language Models
8. Image Generation using GAN's
9. NLP Task – Text Summarization
10. Application Case Study

Text Books

1. "Deep Learning" by Ian Goodfellow, Yoshua Bengio, and Aaron Courville: This comprehensive book covers fundamental concepts in deep learning, including generative models, and provides a solid foundation for understanding the mathematical and theoretical underpinnings of generative AI.
2. "Pattern Recognition and Machine Learning" by Christopher M. Bishop: This textbook offers an in-depth exploration of machine learning principles, including probability and statistics, which are crucial for understanding and developing generative models.
3. "Generative Deep Learning: Teaching Machines to Paint, Write, Compose, and Play" by David Foster: This book specifically focuses on generative models such as GANs and VAEs, and their applications, making it highly relevant for the advanced topics covered in the syllabus.

Reference Books

1. "Probabilistic Graphical Models: Principles and Techniques" by Daphne Koller and Nir Friedman: This book offers a detailed exploration of probabilistic models and inference techniques, providing a strong theoretical background essential for understanding generative models.
2. "Information Theory, Inference, and Learning Algorithms" by David J.C. MacKay: This text provides a comprehensive look at information theory and its applications in inference and learning, which are key components in generative AI.
3. "Bayesian Reasoning and Machine Learning" by David Barber: This book focuses on Bayesian methods, which are fundamental to many generative models, and provides practical insights into applying these techniques in machine learning.





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR

(An Autonomous Institute)

Final Year B.Tech. (Information Technology), Semester-VII

23ITU7E24T- Program Elective-V: Edge Computing

Teaching Scheme

Lectures 2 Hours/week

Practical 2 Hours/week

Credits 3

Examination Scheme

ESE 60 Marks

ISE 40 Marks

ICA 25 Marks

Introduction:

A distributed system is a collection of independent computers, connected by a network, that work together to appear as a single, coherent system to the user. Edge computing Key concepts include mitigating bandwidth congestion, enhancing real-time decision-making, and securing decentralized architectures.

Course Prerequisite:

A basic course on operating systems and algorithms. Foundational knowledge in networking (protocols, latency), and programming (Python/C++)

Course Objectives:

1. To introduce existing knowledge in distributed systems.
2. To review underlying technologies, limitations, and challenges along with performance metrics.
3. To impart the knowledge to log the sensor data and to perform further data analytics.

Course Outcomes:

After completing the course, students will be able to

1. Demonstrate the use of message passing techniques and RPC models to enable communication between distributed processes.
2. Analyze basic synchronization techniques such as logical clocks, mutual exclusion, and coordination algorithms in distributed environments.
3. Describe the role of lightweight container middleware in edge-cloud environments and illustrate how containerization enables efficient deployment at the edge.
4. Apply various application scenarios such as smart cities, healthcare, and autonomous systems to determine the role of Edge computing.



Unit – I	Message Passing in Distributed Systems	3 Hours
Introduction, Desirable features of Good Message-Passing System, Issues in IPC by Message Passing, Synchronization, Buffering, Message Passing Interface, Multi-datagram Messages, Process Addressing, Failure Handling,		
Unit – II	Synchronization in Distributed Systems	6 Hours
Clock Synchronization, Event Ordering, Election algorithms, Process Migration.		
Unit – III	A Lightweight Container Middleware for Edge Cloud Architectures	7 Hours
Docker Orchestration, Docker Orchestration Architecture, Docker Evaluation – Installation, Performance, Power. IoT Integration. Security Management for Edge Cloud Architectures. Security Requirements and Blockchain Principles. A Blockchain-Based Security Architecture, Integrated Blockchain-Based Orchestration		
Unit – IV	Introduction to Fog and Edge Computing	6 Hours
Introduction to Fog Computing, Limitation of Cloud Computing, Differences between Cloud and Fog Computing, Advantages, Business Models, Architecture, Opportunities and Challenges. Edge Computing: Origins of Edge, Edge Helping Low-End IoT Nodes, Architecture, Edge Helping Higher-Capability		
Unit – V	Applications of Fog and Edge Computing	8 Hours
Mobile Offloading, Edge Helping the Cloud, Edge for Augmented Reality, Data Processing on the Edge, Dispersed Learning with Edge/Fog Computing, Video Analytics on the Edge, Edge Computing Applications.		
Text Books		
1. Distributed Operating System Concepts and Design, P. K. Sinha, PHI 2. Fog and Edge computing: Principles and Paradigms by Buyya, Rajkumar, and Satish Narayana Srirama, John Wiley & Sons, USA		
Reference Books		
1. Fog and Edge Computing by Albert Y. Zomaya, Wiley Series 2. Wei Change and Jie Wu, Fog/Edge Computing for Security, Privacy and Applications, Springer, 2021		



**WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR****(An Autonomous Institute)****Final Year B.Tech. (Information Technology), Semester-VII****23ITU7E34T- Program Elective V: Prompt Engineering**

Teaching Scheme		Examination Scheme	
Lectures	2 Hours/week	ESE	60 Marks
Practical	2 Hours/week	ISE	40 Marks
Credits	3	ICA	25 Marks
Introduction:			
Prompt engineering is the art and science of designing, refining, and optimizing inputs (prompts) to guide large language models (LLMs) like GPT-4 or Gemini to generate accurate, relevant, and high-quality outputs. It involves creating precise instructions, providing context, and iterating on language to enhance AI performance for tasks like text generation, coding, and reasoning.			
Course Prerequisite:			
Basic Knowledge of AI, Strong Communication Skills, An understanding of Large Language Models (LLMs).			
Course Objectives:			
1. To introduce the fundamentals of prompt engineering and generative AI systems. 2. To develop the ability to use AI tools and prompt engineering techniques for content generation and personalization. 3. To familiarize students with AI techniques for analysing and processing unstructured text data. 4. To provide exposure to advanced applications of LLMs across different industries. 5. To create awareness about the ethical, societal, environmental, and future implications of generative AI systems.			
Course Outcomes:			
After completing the course, students will be able to 1. Describe the fundamentals of prompt engineering and generative AI systems. 2. Apply AI tools and prompt engineering techniques to generate and personalize content. 3. Analyse unstructured text data using AI techniques. 4. Illustrate advanced applications of LLMs in different industries. 5. Examine the ethical, societal, environmental, and future implications of generative AI systems.			
Unit – I	Introduction to Prompt Engineering		5 Hours
Understanding Prompting and Prompt Techniques, Introducing LLM Prompts, How LLM Prompts Work, Types of LLM Prompts, Components of an LLM Prompts, Using patterns to enhance Prompt			



Effectiveness, Mix and Match – Strategic Combinations for enhanced prompts, Exploring LLM Parameters, How to approach prompt Engineering (Experimentation). The Challenges and Limitations of Using LLM Prompts.		
Unit – II	Generating Text With AI for Content Creation	5 Hours
Using AI for copywriting, Creating Social Media Posts, Writing Video Scripts, Generating Blog posts , Articles and news, Creating Engaging Content with AI, How to use AI for Personalized messaging, Creating Tailored content with AI,		
Unit – III	Basic Prompt Engineering Techniques	4 Hours
Creating and Promoting a Podcast Using ChatGPT and Other Practical Examples: Crafting Podcast questions for celebrity guests, Preparing podcast questions with everyday guests, Identify topics ,ideas and potential guest speakers for your podcast, Using AI to promote a podcast		
Unit – IV	LLM's for Creating Writing	6 Hours
Using AI for creative writing, Using AI to generate fiction, Using AI to write Poetry , Unlocking Insights from Unstructured Text – AI Techniques for Text Analysis: Sentiment Analysis-AI Techniques for emotion detection in Text, Organizing Unstructured data- Using AI for Automated text categorization and data Classification, cleaning up dirty data – How AI identifies and resolves issues in datasets, Making sense of Unstructured data-pattern matching for information extraction.		
Unit – V	Advanced Use Cases for Different Industries	5 Hours
Applications of LLMs in Education and Law, The Rise of AI Pair Programs – Teaming Up with Intelligent Assistants for Better Code, AI for Chatbots, Building Smarter Systems – Advanced LLM Integrations.		
Unit – VI	Ethics, Limitations, and Future Developments	5 Hours
Generative AI – Emerging Issues at the Intersection of Ethics and Innovation: Exploring the ethical challenges of generative AI, Economic Impact Considerations, Environmental Sustainability Issues, Societal risks and reflections, The path forward-solutions and safeguards. Expanding possibilities-innovative prompt engineering applications.		
Text Books		
1. Gilbert Mizrahi, “Unlocking the Secrets of Prompt Engineering: Master the Art of Creative Language Generation to Accelerate Your Journey from Novice to Pro”, January 2024		
Reference Book		
1. James Phoenix, Mike Taylor, “Prompt Engineering for Generative AI”, O'Reilly, May 2024 2. Michael Ferguson, “Prompt Engineering: The Future of Language Generation”, January 2023		



e-Resources

1. Prompt Engineering Guide <https://www.promptingguide.ai>
2. Prompt engineering, <https://developers.openai.com/api/docs/guides/prompt-engineering>



**WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR****(An Autonomous Institute)****Final Year B.Tech. (Information Technology), Semester-VII****23ITU7PR6L: Project Phase I**

Teaching Scheme		Examination Scheme	
Lectures	-	ESE	100 Marks
Practical	8 Hours/week	ICA	50 Marks
Credits	4		
Introduction:			
This Phase deals with conceptualising an idea. This idea is then elaborated to a design which is performed taking into consideration the requirements of the problem defined.			
Course Prerequisite:			
Methodology of idea conceptualization, designing methods like conventional model creation or UML.			
Course Objectives:			
1. To guide students to explore research areas and to undertake literature survey. 2. To identify & formulate a realistic problem statement. 3. To define the problem of the proposed research work 4. To follow an appropriate designing technique for further development of project. 5. To prepare to work in a team and to understand importance of teamwork. 6. To develop soft skills including presentation and technical writing			
Course Outcomes:			
At the end of the course students will be able to 1. Explore research areas, conduct literature survey and formulate a problem statement catering societal/professional need. 2. Able to identify and formulate project problem statement and select an appropriate design with due consideration for society. 3. Carry out impact analysis for environment and sustainability concern. 4. Prepare Software requirement specification (SRS) and design document using software engineering techniques and modern tools.			



5. Engage in team work and communicate effectively while observing professional ethics.
6. Inculcate the habit of self-study to become a lifelong learner.

Guidelines :

The objective of Project Phase–I is to enable students to identify, analyze, and design a solution for a real-world engineering problem. Students are expected to perform detailed study, requirement analysis, literature review, and system design under the guidance of the project supervisor.

1. Students shall finalize the project topic after obtaining approval from the project guide and department.
2. Each project group shall submit: Project synopsis, Presentation/Seminar, and Project Phase–I report.
3. The project synopsis should ideally contain: Title of the project, Abstract, Problem statement, Objectives, Scope of the project, Literature survey/review, Proposed methodology, System architecture/design, Hardware/software requirements, and Expected outcomes.
4. Students shall perform detailed study and analysis of the selected problem domain.
5. The project design should include: System flow, Architectural diagrams, Database design (if applicable), UML diagrams/DFD/ER diagrams, Algorithm or workflow design, and Module identification.
6. Students shall identify suitable tools, technologies, frameworks, programming languages, and platforms required for implementation.
7. A seminar/presentation on the proposed system design and methodology shall be conducted before the departmental review committee.
8. The Phase–I report should include: Introduction, Literature review, Problem definition, Requirement analysis, Proposed system and methodology, System design, Feasibility analysis, Work plan and timeline, and References.
9. Students are expected to demonstrate preliminary implementation, prototype design, simulation results, or proof-of-concept wherever applicable.
10. Proper documentation standards, plagiarism policy, and ethical practices shall be followed throughout the project work.
11. Continuous assessment shall be carried out based on: Literature survey quality, Problem understanding, Innovation and feasibility, System design, Presentation skills, and Documentation quality.





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)
Final Year B. Tech. (All Branches), Semester-VIII

23CMU8R11T: Research Methodology and Intellectual Property Rights

Teaching Scheme		Examination Scheme	
Lectures	-	ESE	50 Marks
Credits	2		

Introduction

Research Methodology provides a systematic framework for identifying, analyzing, and solving technical problems through evidence-based inquiry. Intellectual Property Rights (IPR) establish the legal safeguards necessary to protect these engineering innovations, ensuring that creators maintain control and commercial benefits over their inventions. Together, they equip students with the tools to conduct ethical research and navigate the global patent and copyright landscape.

Course Prerequisites:

Basics of Engineering and Statistics

Course Objectives:

This course aims to:

1. Impart knowledge on research types, processes, design, and literature review techniques.
2. Develop skills in data collection, sampling, and statistical analysis of research data.
3. Enable preparation of research reports with standard citations and inculcate research ethics.
4. Provide understanding of IPR types, legal frameworks, patent search, and filing process.

Course Outcomes:

At the end of this course, students will be able to:

1. Demonstrate the fundamental concepts of research design, its types, processes, and systematic literature review techniques.
2. Analyze research data by using appropriate data collection methods, sampling techniques, and statistical tools.
3. Prepare research reports using standard citation formats and demonstrate ethical responsibilities in research conduct.
4. Classify IPR types and legal frameworks, and apply knowledge to patent search and filing process.

Unit – I	Introduction to Research	
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Meaning & objectives of research, Types of research: basic, applied, quantitative, qualitative, Steps in research process, identifying research problems, Formulating hypotheses and objectives



Unit – II	Literature Review & Research Design	
Literature search strategies, IEEE Xplore, Scopus, Google Scholar, Springer, Web of Science; Research gap analysis, Research design: exploratory, descriptive, experimental, Sampling techniques		
Unit – III	Data Collection & Analysis	
Primary vs secondary data, Tools of data collection (questionnaires, surveys, interviews), Basics of statistical analysis: Mean, Median, Mode, Standard deviation, correlation; Use of software (Excel, SPSS), Data interpretation and presentation.		
Unit – IV	Technical Communication & Ethics in Research	
Technical writing framework, Research paper structure: Abstract, Introduction, Methodology, Results, Discussion, Conclusion; Referencing and citations (APA/IEEE formats), Ethical considerations & plagiarism, Use of plagiarism checking tools		
Unit – V	Introduction to Intellectual Property Rights	
IPR definitions & relevance to engineering, Patents: types, patentable subject matter; Copyright: scope, ownership, duration; Trademarks, Trade Secrets, Industrial Designs		
Unit – VI	Patent Rules, Filing, and IP Management	
Patent search strategies, Patent drafting basics, Patent filing procedure in India and abroad, Patent Cooperation Treaty (PCT) overview, IPR protection strategy for products and software, Technology transfer, Licensing		
Text Books		
1. Research Methodology, A Practical and Scientific Approach - Vinayak Bairagi, Mousami V. Munot, Chapman and Hall 2. Research Methodology: C.R. Kothari, New Age International Publications 3. Research methodology: A step-by-step guide for beginners - Kumar, R. (2018) 4. Intellectual Property Rights: P. Narayanan, S. Chand Publication		
Reference Books		
1. “Research Methodology: Methods & Techniques” — Ranjit Kumar, Sage Publications 2. Intellectual Property Rights under WTO, T. Ramappa 3. Intellectual Property Handbook, WIPO (World Intellectual Property Organization, free PDF).: https://tind.wipo.int/record/28661/files/wipo_pub_489.pdf		
Relevant Videos on Youtube related to previous NPTEL Courses		
1. Research Methodology — NPTEL course (Videos Available on Youtube) https://www.youtube.com/watch?v=E2gGF1rburw&list=PLyqSpQzTE6M8F_P8lgjvmqiDEo_FGLzG4h&index=1 2. Intellectual Property – NPTEL course (Videos Available on Youtube) https://www.youtube.com/watch?v=a-		



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Important Websites / E-Resources

1. IEEE Xplore Digital Library: <https://india.ieee.org/ieee-xplore-digital-library/>
2. Google Scholar: <https://scholar.google.com/>

Patent Databases (url)

1. WIPO (World Int'l Patents): <https://www.wipo.int/en/web/patents>
2. Indian Patent Advanced Search System (InPASS): <https://iprarch.ipindia.gov.in/publicsearch>





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR

(An Autonomous Institute)

Final Year B.Tech. (Information Technology), Semester-VIII

23ITU8E12T - Program Elective VI: Software Testing and Quality Assurance

Teaching Scheme		Examination Scheme	
Lectures		ESE	50 Marks
Credit	2		
Introduction:			
Software Testing and Automation are distinct yet complementary processes ensuring software excellence. QA is a proactive approach focused on improving development processes to prevent defects, while software testing is a reactive, tactical process aimed at identifying and documenting bugs in the final product.			
Course Prerequisite:			
Require core knowledge in software fundamentals, Software Development Life Cycle, specific technical skills.			
Course Objectives:			
1. To apply the testing strategies and methodologies in projects. 2. To understand test management strategies and tools for testing. 3. To apply software test automation and explore various tools			
Course Outcomes:			
After completing the course, students will be able to 1. Design test cases suitable for software development for different domains. 2. Prepare a test planning based on the document. 3. Implementing a test plan and design test cases. 4. Use automatic testing tools.			
Unit – I	Introduction		
Testing as an Engineering Activity - Testing as a Process - Testing Maturity Model - Testing axioms - Basic definitions - Software Testing Principles - The Tester 's Role in a Software Development Organization - Origins of Defects - Cost of defects - Defect Classes - The Defect Repository and Test Design - Defect Examples - Developer/Tester Support of Developing a Defect Repository.			



Unit – II	Test Case Design Strategies	
Test case Design Strategies - Using Black Box Approach to Test Case Design - Boundary Value Analysis - Equivalence Class Partitioning - State based testing - Cause-effect graphing - Compatibility testing - user documentation testing - domain testing - Random Testing - Requirements based testing - Using White Box Approach to Test design - Test Adequacy Criteria - static testing vs. structural testing - code functional testing - Coverage and Control Flow Graphs - Covering Code Logic - Paths - code complexity testing - Additional White box testing approaches - Evaluating Test Adequacy Criteria.		
Unit – III	Levels of Testing	
People and organizational issues in testing - Organization structures for testing teams - testing services - Test Planning - Test Plan Components - Test Plan Attachments - Locating Test Items - test management - test process - Reporting Test Results - Introducing the test specialist - Skills needed by a test specialist - Building a Testing Group - The Structure of Testing Group - The Technical Training Program.		
Unit – IV	Test Management	
People and organizational issues in testing - Organization structures for testing teams - testing services - Test Planning - Test Plan Components - Test Plan Attachments - Locating Test Items - test management - test process - Reporting Test Results - Introducing the test specialist - Skills needed by a test specialist - Building a Testing Group - The Structure of Testing Group - The Technical Training Program.		
Unit – V	Test Automation	
Software test automation - skills needed for automation - scope of automation - design and architecture for automation - requirements for a test tool - challenges in automation - Test metrics and measurements – project-progress and productivity metrics. Automation Case Studies on -Selenium Tools. Emerging AI Testing Tools like Playwright, basic framework design and CI/CD integration.		
Text Books		
1. “Software Testing – Principles and Practices”, Srinivasan Desikan, Gopalaswamy Ramesh, Pearson Education 2. “Software Testing”, Second Edition, Ron Patton, Sams Publishing, Pearson Education		
Reference Books		
1. “Practical Software Testing”, Ilene Burnstein, Springer International Edition 2. “Software Testing in the Real World – Improving the Process”, Edward Kit, Pearson Education		



**WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR****(An Autonomous Institute)****Final Year B.Tech. (Information Technology), Semester-VIII****23ITU8E22T: Program Elective Course VI-Augmented Reality and Virtual Reality**

Teaching Scheme		Examination Scheme	
Lectures	-	ESE	50 Marks
Credits	2	ICA	-
Introduction:			
This course will introduce Virtual Reality and Augmented Reality, with an emphasis on understanding virtual and augmented reality concepts, theoretical foundations, interaction techniques, and application areas.			
Course Prerequisite:			
Computer Graphics, Basics of 3D Geometry			
Course Objectives:			
1. To understand the fundamental concepts and framework of Virtual Reality (VR). 2. To analyze the hardware and software requirements of Virtual Reality and Augmented Reality systems. 3. To apply different 3D interaction techniques used in VR environments. 4. To understand effective 3D user interfaces for VR applications. 5. To explore various applications of Virtual Reality and Augmented Reality in different fields.			
Course Outcomes:			
After completing the course, students will be able to: 1. Demonstrate the basic concept and framework of virtual reality 2. Analyse the hardware and software requirements for VR and AR 3. Use the different 3D interaction techniques. 4. Describe 3D interfaces in VR 5. Explore VR and AR applications			



Unit – I	Virtual Reality and Hardware Technologies for 3D User Interfaces	
Virtual reality and virtual environments: the historical development of VR, scientific landmarks, computer graphics, real-time computer graphics, virtual environments, requirements for VR, benefits of virtual reality. Hardware technologies for 3D user interfaces: visual displays, auditory displays, haptic displays, choosing output devices for 3D user interfaces.		
Unit – II	3D User Interface Input Hardware and Software Technologies	
3D user interface input hardware: input device characteristics, desktop input devices, tracking devices, 3D mice, special purpose input devices, direct human input, home-brewed input devices, choosing input devices for 3D interfaces. Software technologies: database – world space, world coordinate, world environment, objects – geometry, position/orientation, hierarchy, bounding volume, scripts and other attributes, VR environment – VR database, tessellated data, LODs, cullers and occluders, lights and cameras, scripts, interaction – simple, feedback, graphical user interface, control panel, 2D controls, hardware controls, room/stage/area descriptions, world authoring and playback, VR toolkits, available software in the market.		
Unit – III	3D Interaction Techniques in Virtual Reality	
3D interaction techniques: 3D manipulation tasks, manipulation techniques and input devices, interaction techniques for 3D manipulation, design guidelines – 3D travel tasks, travel techniques, theoretical foundations of way finding, user-centered way finding support, environment-centered way finding support, evaluating way finding aids, system control, classification, graphical menus, voice commands, gestural commands, tools, multimodal system control techniques, case study: mixing system control methods, symbolic input tasks, symbolic input techniques, beyond text and number entry.		
Unit – IV	Designing and Advances in 3D User Interfaces	
Designing and developing 3D user interfaces: strategies for designing and developing guidelines and evaluation. Advances in 3D user interfaces: 3D user interfaces for the real world, AR interfaces as 3D data browsers, 3D augmented reality interfaces, augmented surfaces and tangible interfaces, agents in AR, transitional AR–VR interfaces – the future of 3D user interfaces, questions of 3D UI technology, 3D UI design and development, 3D UI evaluation and other issues.		
Unit – V	Virtual Reality Applications	



Virtual reality applications: engineering, architecture, education, medicine, entertainment, science, training.		
Text Books		
1. Paul Mealy, Virtual & Augmented Reality for Dummies, John Wiley & Sons. 2. Alan B Craig, William R Sherman and Jeffrey D Will, —Developing Virtual Reality 3. Applications: Foundations of Effective Design, Morgan Kaufmann.		
Reference Books		
1. Jan Erik Solem, Programming Computer Vision with Python, ShroffPublisher/O'Reilly Publisher 2. Gerard Jounghyun Kim, —Designing Virtual Systems: The Structured Approach 3. Doug A Bowman, Ernest Kuijff, Joseph J LaViola, Jr and Ivan Poupyrev, —3D User Interfaces, Theory and Practice, Addison Wesley, USA		





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR

(An Autonomous Institute)

Final Year B.Tech. (Information Technology), Semester-VIII

23ITU8IN3L: Internship

Teaching Scheme		Examination Scheme	
Lectures	-	ESE	50 Marks
Practical	20 Hours/week	ICA	100 Marks
Credits	10		
Introduction:			
The purpose of the internship program is to provide students with practical experience in a standard work environment and help them develop the skills necessary for a successful professional career. Internship offers an opportunity to gain knowledge and skills through planned work experience in the student's chosen career field. It bridges the gap between classroom learning and workplace application by exposing students to industry practices, professional responsibilities, and career-related competencies. Internships also help students explore career paths, build confidence, and meet core learning outcomes through real-world experience.			
Course Prerequisite:			
Students should have foundational knowledge of programming, data structures, database management systems, computer networks, software engineering, and basic analytical skills required for carrying out internship work in the Information Technology field.			
Course Objectives:			
1. To give exposure to the industrial environment and create competent professionals for the industry. 2. To provide possible opportunities to learn, understand, and sharpen the real-time technical/managerial skills. 3. To familiarize oneself with processes, products, software, and their applications along with relevant aspects of quality control. 4. To understand various approaches towards problem-solving. 5. To expose students to the engineer's responsibilities and ethics. 6. To understand the social, economic, and administrative considerations that influence the working environment of industrial organizations.			
Course Outcomes:			
At the end of the course, students will be able to 1. Develop professional competence through internship experience. 2. Apply academic knowledge in a personal and professional environment. 3. Build a professional network.			



4. Apply professional and societal ethics in day-to-day life.
5. Become a responsible professional having social, economic, and administrative considerations.
6. Make own career goals and personal aspirations.

Guidelines:

1. Students may undergo internships in small, medium, or large-scale industries to gain industry exposure and enhance their professional readiness.
2. Students shall undergo the internship for a minimum period of three months during the semester.
3. The evaluation of internship work shall be carried out by the program head, cell in-charge, project coordinator, project guide, faculty mentor, and industry mentor, wherever applicable.

The industry shall appoint a supervisor to monitor and assess the student's performance during the internship and submit the assessment report to the departmental coordinator for ICA evaluation.

4. Every intern shall submit a weekly progress report to the internal guide and shall maintain communication with the internal guide at least once every fortnight.
5. Students shall maintain an internship diary/workbook to record daily activities, observations, information gathered, learning outcomes, and suggestions, if any.
6. The student shall deliver a seminar based on the internship training report every month before an expert committee constituted by the department, as per institute norms, as part of ICA.
7. The internship report shall be prepared in the prescribed format. and shall include all the recommended sections relevant to the internship work carried out.
8. The internship report shall be accepted for final evaluation only if the student regularly reports the progress of work to the project guide and undergoes periodic evaluation by the internal examiners at the institutional level.



**WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR****(An Autonomous Institute)****Final Year B.Tech. (Information Technology), Semester-VIII****23ITU8PR4L: Project Phase II**

Teaching Scheme		Examination Scheme	
Lectures	-	ESE	50 Marks
Practical	4 Hours/week	ICA	100 Marks
Credits	2		

Introduction:

This Phase is the implementation phase where the design created in the phase I is brought into reality as a solution to a problem selected.

Course Prerequisite:

Basics of programming, design and computer languages.

Course Objectives:

- 1.To study methods to analyse technological alternatives for developing IT solution with relevance to environment and sustainability.
2. To explore state-of-art tools and FOSS alternatives to develop solutions meeting societal and professional needs.
3. To enable development of a system through Software Development Life Cycle.
4. To get acquainted to work in teams observing professional ethics.
5. To enable effective written and oral communication for presentation.
6. To enable self-study and lifelong learning

Course Outcomes:

At the end of the course students will be able to

1. Analyse technological alternatives for developing IT solution with relevance to environment and sustainability.
2. Explore state-of-art tools and FOSS alternatives to develop solutions meeting societal and professional needs.
3. Design and develop solution to the problem through Software Development Life Cycle.



4. Demonstrate ability to engage in teamwork while observing professional ethics.
5. Communicate project work in writing and oral presentation.
6. Inculcate the habit of self-study to become a lifelong learner.

The objective of Project Phase–II is to extend the investigative study carried out in Project Phase–I into full-scale implementation, experimentation, testing, analysis, and validation of the proposed solution.

1. Students shall continue the project work initiated in Project Phase–I under the guidance of the approved supervisor.
2. The project may involve: Practical implementation, Experimental work, Simulation/modeling, Hardware/software development, and Analytical or research-oriented study.
3. Students shall perform an in-depth study and refinement of the methodology proposed during Phase–I.
4. The assignment should normally include: Detailed analysis and modeling of the problem, System implementation and development, Algorithm design/coding/testing, Simulation or experimentation, Performance evaluation and result analysis, and Validation of outcomes.
5. Students shall finalize and implement the complete system/product/process.
6. Proper testing and verification procedures should be carried out and documented.
7. The Phase–II report should include: Introduction and objectives, Review of Phase–I work, Detailed methodology, System implementation, Experimental setup, Results and discussion, Performance analysis, Conclusion, Future scope, and References.
8. Students are encouraged to publish research papers, present papers in conferences/seminars, participate in project competitions/hackathons, and file patents/copyrights if applicable.
9. Final project demonstration and oral presentation shall be conducted before the departmental evaluation committee.
10. Evaluation shall be based on: Technical implementation, Innovation and originality, Quality of results and analysis, Teamwork and work ethics, Documentation quality, and Presentation and demonstration skills.
11. Students shall maintain professional engineering practices, proper version control, documentation standards, and ethical conduct throughout the project work.
12. The final project report must be prepared in the prescribed standard format provided by the department.

